MONASH University

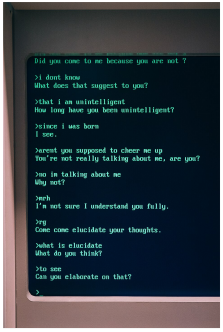
Can Machines Think?

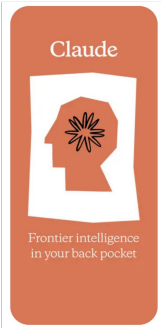
Week 1 — FIT1061 Introduction to AI

Julian Garcia

FIT1061 — Introduction to Artificial Intelligence

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MILESTONE

ELIZA to LLM's

1966 — 2024

The same question. Different machines.

Try ELIZA: elizaemulator.com

Poll

Is ChatGPT intelligent?

- **Yes**
- **No**
- **It depends what “intelligent” means**

Vote now — scan the code for your campus, *or* type the address.



Melbourne

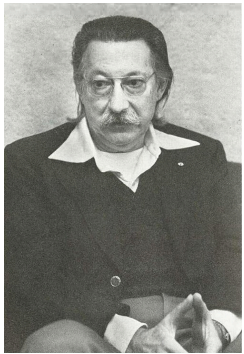
pollev.com/juliangarcia



Malaysia

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What changed?

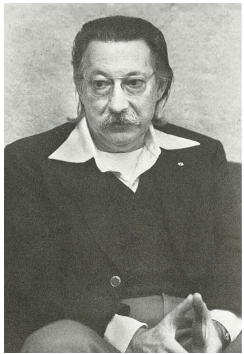


Joseph Weizenbaum at MIT, c. 1966.

ELIZA — 200 lines of LISP.

- Both can hold a conversation.
- Both have fooled people.

What changed?



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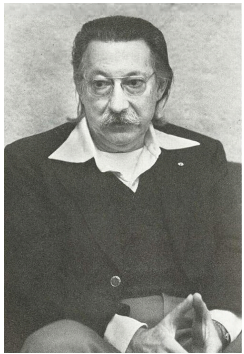
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ELIZA: pattern-matching script.

- ~200 lines of code.

What changed?



Joseph Weizenbaum at MIT, c. 1966.

ELIZA — 200 lines of LISP.

- Both can hold a conversation.
- Both have fooled people.

ELIZA: pattern-matching script.

- ~200 lines of code.

Modern LLMs (ChatGPT, Claude, Gemini, …):

- **100B+ parameters**
- Trillions of training tokens
- ~$100M compute

But can either of them think?

Today

Can machines think?

Is it AI?

Three questions

How the unit works

Can machines think?

Is it AI?

Three questions

How the unit works

Can machines think?

The imitation game



Alan Turing, 1950



The medium: teletype

“I propose to consider the question, ‘Can machines think?’ … I shall replace the question by another … The new form of the problem can be described in terms of a game which we call the ‘imitation game’.”

— Alan Turing, 1950

The imitation game



Alan Turing, 1950



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The game: a judge texts with a human and a machine.
Text only. Five minutes. **Which is the machine?**

The imitation game



Alan Turing, 1950



The medium: teletype

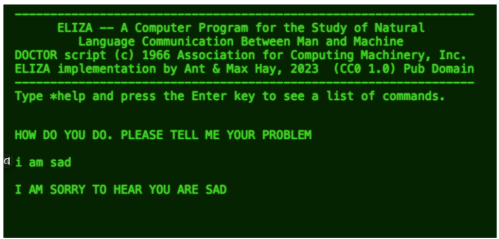
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— Alan Turing, 1950

The game: a judge texts with a human and a machine.
Text only. Five minutes. **Which is the machine?**

The test: if the judge can’t reliably tell,
that is what we will call *thinking*.

Fooling people, 1966 – 2014



ELIZA, 1966.
Pattern-matching, no understanding.
Patients asked to be left alone with it.



Goostman, 2014.
Posed as a 13-year-old Ukrainian.
Fooled 33% of judges in 5-minute chats.
Headlines: “Turing Test passed”.

Would ChatGPT pass the Turing test?



A modern chat UI

- **2022:** GPT-3.5 holds a 30-minute conversation indistinguishable from a thoughtful human for most users.
- **2023:** GPT-4 passes the bar exam, medical licensing, university-entry standardised tests.
- **2024–25:** Reasoning models (o1, DeepSeek R1) solve Olympic-level maths problems.

Would ChatGPT pass the Turing test?



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- **2024–25:** Reasoning models (o1, DeepSeek R1) solve Olympic-level maths problems.

"Intelligence is a suitcase word. It contains many different meanings, and the suitcase is too small."

— Melanie Mitchell, 2019

Turing's question — "can machines think?" — became "what does *thinking* mean?"

Can machines think?

Is it AI?

Three questions

How the unit works

Is it AI?

Mechanical thermostat



Bimetallic coil tilts a mercury switch. No code.

AI? – Not AI?

Nest learning thermostat



Adjusts the schedule from your behaviour.

AI? – Not AI?

Nest learning thermostat

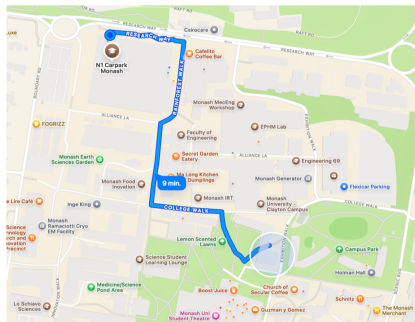


Adjusts the schedule from your behaviour.

AI? – Not AI?

The “learn from data” machinery: weeks 10–11.

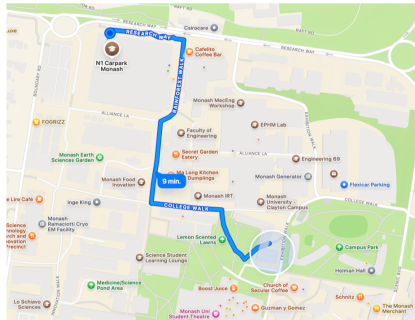
Google Maps routing



Finds the shortest path. Solved problem since 1959.

AI? – Not AI?

Google Maps routing

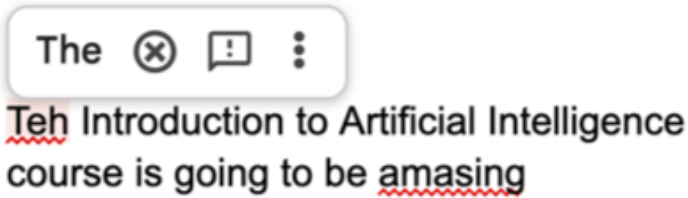


Finds the shortest path. Solved problem since 1959.

AI? – Not AI?

Shortest path on a graph. Same idea drives every search algorithm in weeks 2–5.

Word spellcheck



Word not in dictionary: suggest the closest match by edit distance.

AI? – Not AI?

Spam filter



Filters mail into “inbox” vs “spam”.

AI? – Not AI?

Spam filter



Filters mail into “inbox” vs “spam”.

AI? – Not AI?

Estimates $P(\text{spam} \mid \text{words})$ from past examples. We build one in week 7.

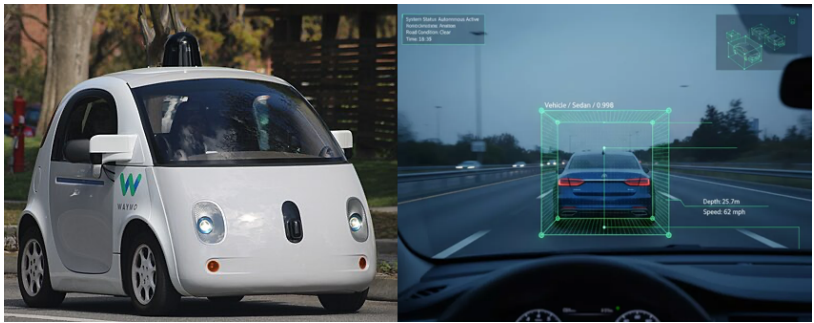
Roomba



Maps the room. Plans a path. Doesn't learn.

AI? – Not AI?

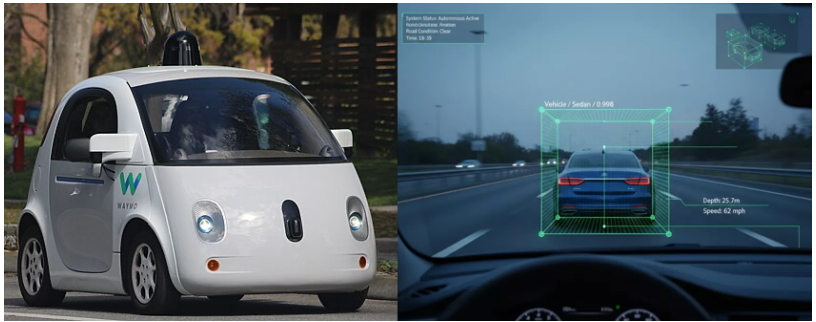
Tesla Autopilot



Camera + deep network steers the car.

AI? – Not AI?

Tesla Autopilot

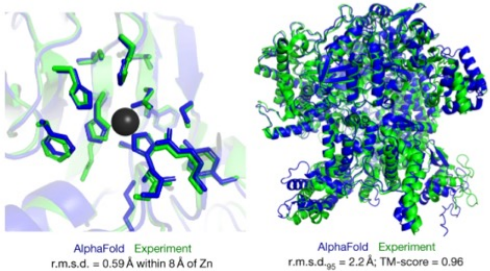


Camera + deep network steers the car.

AI? – Not AI?

A neural network learns to recognise the road. Computer vision: week 11.

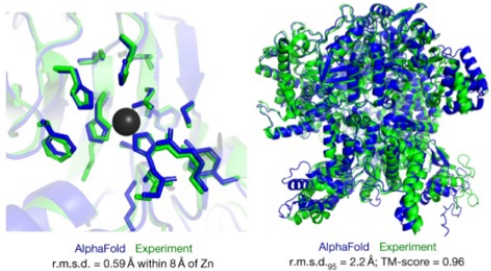
AlphaFold (Nobel Prize, 2024)



Predicts the 3D shape of any protein. **Won the Nobel**.

AI? – Not AI?

AlphaFold (Nobel Prize, 2024)

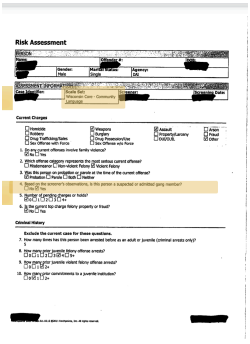


Predicts the 3D shape of any protein. **Won the Nobel**.

AI? – Not AI?

A neural network trained by nudging weights downhill, billions of times. Week 10.

COMPAS recidivism score



Risk Assessment

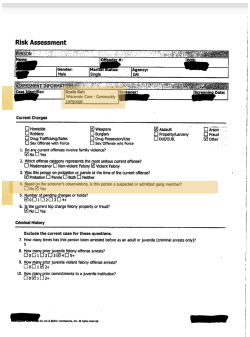
Current Charges

Criminal History

137 inputs → risk of reoffending. Used in US courts.

AI? – Not AI?

COMPAS recidivism score



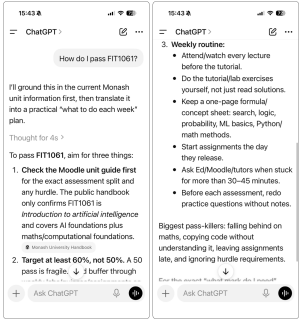
Risk Assessment

137 inputs → risk of reoffending. Used in US courts.

AI? – Not AI?

Classifier trained on past cases. We dissect this exact case in week 9.

LLM writing this lecture



Generates the words you're reading.

AI? – Not AI?

A provisional AI definition

“AI is the science of making machines do things that would require intelligence if done by humans.”

— John McCarthy, 1956

But...

“As soon as it works, no one calls it AI anymore.”

— John McCarthy — who coined “artificial intelligence”, 1956

But...

“As soon as it works, no one calls it AI anymore.”

— John McCarthy — who coined “artificial intelligence”, 1956

“AI is whatever hasn’t been done yet.”

— Arvind Narayanan & Sayash Kapoor, *AI Snake Oil* (2024)

We will not use **AI** as a definition in this unit. We will use **algorithms**.

Can machines think?

Is it AI?

Three questions

How the unit works

Three questions

Three questions. Twelve weeks.

Block A	Block B	Block C
How machines search	*Decisions under uncertainty*	*How machines learn*
W2 BFS / DFS	W6 Bayes	W10 gradient descent
W3 greedy	W7 Naive Bayes	W11 convolution
W4 hill climbing	W8 perceptron	W12 deployment
W5 consolidation	W9 fairness	

One leading example per block. Then we'll see how LLMs combine all three.

How do machines search?



AlphaGo (DeepMind, 2016).
Beat Lee Sedol 4–1 at Go.
Move 37 stunned commentators — no human had played it.

The recipe: search + learning

- A neural network proposes promising moves.
- A tree search explores their consequences.
- Self-play feeds the network better data.

What you'll learn in Block A

- Breadth-first & depth-first search
- Greedy & A* heuristic search
- Hill climbing
- Minimax (game trees)

Same recipe behind AlphaProof (IMO silver 2024), Gemini Deep Think (IMO gold 2025), and every agent that plans — Cursor, Claude Code, ChatGPT Agents.

How do machines decide under uncertainty?



Watson (IBM, 2011).
Beat Jeopardy! champions Brad Rutter and Ken Jennings. Probability over millions of facts.

COMPAS (US courts, 2016).
Same machinery, higher stakes — bias hidden in the data (we'll meet this case in a few weeks).

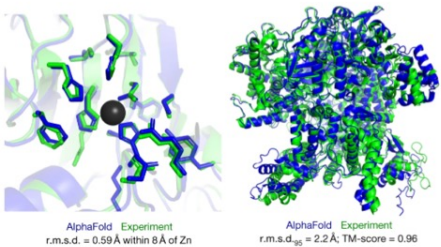
The recipe: probability over evidence

- Estimate Pr(outcome | data) from past examples.
- Combine features into a single decision.
- Audit who the model gets right — and wrong.

What you'll learn in Block B

- Probability & Bayes' rule
- Naive Bayes (*the spam-filter algorithm*)
- Weighted sums — unit cell of every neural net
- Fairness metrics

How do machines learn?



AlphaFold (DeepMind, 2020–24).
Predicts the 3D shape of any protein.
A 50-year-old open problem.
Hassabis & Jumper: **2024 Nobel Prize in Chemistry**.

The recipe: gradient descent at scale

- Start with millions of labelled examples.
- Define a loss — how wrong is each prediction?
- Nudge the weights downhill, billions of times.

What you'll learn in Block C

- Gradient descent (1D and 2D)
- Perceptron training
- Convolution — how machines see
- Deployment & model cards

Same recipe behind ChatGPT, Tesla Autopilot, weather forecasting (GraphCast), and AlphaFold.

And LLMs?

LLM's use **all three blocks**.

- **Block A (search):** agents that wrap LLMs search over actions — Cursor, Claude Code. Reasoning models search over chains of thought.
- **Block B (probability):** next-token prediction *is* a conditional probability. The perceptron is the transformer's unit cell.
- **Block C (learning):** gradient descent trained the weights. RLHF polished the behaviour.

Can machines think?

Is it AI?

Three questions

How the unit works

How the unit works

12 weeks, 12 milestones

	Block A · Search				Block B · Probability				Block C · Learning		
W1 ELIZA → ChatGPT	**W2** Deep Blue	**W3** AlphaGo Move 37	**W4** SW Air. collapse	**W5** Haiti routing	**W6** Watson Jeopardy!	**W7** COMPAS (2016)	**W8** Perceptron (1958)	**W9** Gender Shades	**W10** AlexNet (2012)	**W11** Computer vision	**W12** AI safety

The milestone is *why* we built the algorithm.

You pick your level. You can change it any time before Week 13.

The ladder

HD 80+

Comprehensive project: define → build → defend in an interview.

Distinction 70–79

Design your own solutions. A*, perceptron, tournament agent, video argument.

Credit 60–69

Less scaffolding. Fairness audit. Computer vision. Deployment.

Pass 50–59

Implement the week's algorithm with scaffolding. Engage with the milestone.

Raise or lower your target any time before Week 13 — one paragraph in your Learning Plan.

Workload & redos

Tier	Workload
Pass	~8 h / week
Credit	~10–12 h / week
Distinction	~12 h / week
HD	~12–15 h / week

Roughly **12 h / week** is the Monash norm; your target moves it up or down. Includes workshop, lab and self-study — time spent does not guarantee a grade.

Each task

Complete — task accepted.
Fix and Resubmit — feedback given. Try again.

As many attempts as you need, in good faith. No caps at any tier — but your tutor may pause resubmissions that stop engaging with feedback.

Hurdle Test – Week 10

30–40 min. Paper. Short answer + hand-computation.
No code. No devices.

What it covers

Hand-traces of search algorithms. Hand-computed Bayes' rule. Confusion-matrix arithmetic. Gradient-descent step calculations.

All by-hand sub-tasks you have already done in your portfolio.

Resit: Week 12. | **Supplementary:** Week 14 (capped at Pass).

The unit cannot be passed without passing the hurdle.

AI use scales with tier

Tier	OK	Not OK
Pass	Ask concept questions (“what is a queue?”) — **cross-check** against the reading or your tutor	Pasting AI code; debugging your code; trusting AI as your only source
Credit	Debug your own code	AI-drafted submissions
Distinction	Pair-program; brainstorm	Code/arguments you can’t trace in discussion
HD	Critical evaluation expected	Uncritical reliance

All AI use cited (tool, prompt, date) — every tier.

Monash AI-acknowledgement guidelines: monash.edu/learning-with-ai

Kialo — three weeks of structured debate

Kialo a debate app | kialo-edu.com

In three weeks: **a prompt → your response → a counter-argument → your reply**.

The debates feed those weeks' reflective responses.

Three debates, across three topics:

Can a chess machine's win tell us anything about intelligence? (Deep Blue)
Do risk scores belong in bail decisions? (COMPAS)
Is public face-recognition acceptable when it's accurate? (face recognition)

The debates run in weeks 2, 7, and 11. Citing week 2 is optional; weeks 7 and 11 expect it.

This week's lab

This week

- Install Python, Jupyter, and an editor of your choice.
- Log in to **OnTrack** — this is where you submit every task.
- Draft your **Learning Plan** — which tier are you aiming for, and **why**?
- Work through the **Python warm-up** — translate last year's C++ into Python.

Your first task is due end of Week 1. Bring questions to lab.

Poll — vote again

Is ChatGPT intelligent?

- **Yes**
- **No**
- **It depends what “intelligent” means**

Vote now — scan the code for your campus, *or* type the address.

Melbourne
pollev.com/juliangarcia

Malaysia
pollev.com/onghueyfang767

Discussion

Discussion

"I had not realised that extremely short exposures to a relatively simple computer program could induce powerful delusional thinking in quite normal people."
— Joseph Weizenbaum, *Computer Power and Human Reason* (1976)

Weizenbaum built ELIZA in 1966. He spent the rest of his life warning about it. Sixty years on, the program has gotten much bigger. The delusion has not changed shape.

Where do you land?

Is ChatGPT intelligent? Defend your answer.